MINHAZUL ISLAM

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School of Software Technology, Zhejiang University, Ningbo, Zhejiang, China

Research Interests

Human-centered AI and human–AI interaction; large language model systems for high-stakes and sensitive domains; AI-mediated communication and computer-supported peer support for mental health; trustworthy and verifiable LLM output (grounding, citation enforcement, hallucination detection, and evaluation methodology); affective and physiological computing for wellbeing. My broader goal is to study how language-model systems can be made accountable enough to be trusted in domains where an unsupported claim carries real cost — clinical peer support, health communication, and expert decision-making.

Education

- **Zhejiang University** *Sept. 2023 – June 2026 (expected)*
  M.Eng. in Industrial Design Engineering Ningbo, China
  - **Thesis:** *BepsBot: A Dual-Mode Writing Assistant for Online Bipolar Disorder Peer Support Communities.*
- **Yunnan University** *Sept. 2019 – June 2023*
  B.Eng. in Computer Science and Technology Kunming, China
  - **Thesis:** Product Recommendation System Based on a Collaborative Filtering Algorithm.

Publications & Presentations

Peer-Reviewed Conference Papers

[C.1] **Minhazul Islam**, Mengru Xue, and Tasnim Afra (2025). *Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework*. In *Entertainment Computing — ICEC 2025, IFIP TC 14 Workshops*, Tokyo, Japan, pp. 61–75. doi:10.1007/978-3-032-02534-0_8

[C.2] Tasnim Afra, Mengru Xue, and **Minhazul Islam** (2025). *Enhancing Biofeedback Interventions for Depression and Anxiety Through Entertainment Computing: A Systematic Review*. In *Entertainment Computing — ICEC 2025, IFIP TC 14 Workshops*, Tokyo, Japan, pp. 37–51. doi:10.1007/978-3-032-02534-0_6

Conference Presentations & Non-Archival Papers

[P.1] **Minhazul Islam**, Mengru Xue, and Tasnim Afra (2026). *BEPSBot: Draft-Grounded AI Writing Assistance for Bipolar-Related Online Peer Support*. Oral presentation, *HHME 2026 PCC*, Zhejiang University, China. [conference page]

[P.2] Tasnim Afra, Mengru Xue, and **Minhazul Islam** (2026). *How Should Voice Agents Respond to Anger and Sadness? A Comparison of Empathic Response, Affect-Neutral Acknowledgment, and Cognitive Reappraisal*. Oral presentation, *HHME 2026 PCC*, Zhejiang University, China. [conference page]

Research Experience

- **Zhejiang University — Graduate Research Assistant** *Sept. 2023 – 2026*
  Advisor: Prof. Mengru Xue Ningbo, China
  - **AI-mediated peer support for bipolar disorder (BEPSBot).** Designed and built a draft-grounded writing-assistance system that helps peer supporters compose responses in online bipolar communities, constraining generative suggestions to the supporter's own draft to preserve authorial voice and reduce fabricated advice.

- Constructed the underlying corpus and feature pipeline: large-scale collection of bipolar-related peer-support threads via the Pushshift API, linguistic and sentiment feature extraction (LIWC, TextBlob), and supervised comment-quality classifiers (SVM, Random Forest, XGBoost) benchmarked against transformer baselines (BERT, Sentence Transformers).
- Implemented semantic retrieval over the corpus (Elasticsearch + sentence embeddings) so that suggestions are grounded in comparable prior peer responses rather than generated unconditionally.
- **Systematic review methodology.** Led a systematic framework study of psychologist-applied biomarkers in bipolar disorder [C.1]: defined the search and screening protocol, developed the coding scheme, and synthesized findings into a framework linking biomarker modalities to applied psychological practice. Contributed screening, coding, and analysis to a companion review on biofeedback interventions for depression and anxiety [C.2].
- **Research through design.** Designed and evaluated SPROUT, a shape-changing meditative artifact targeting fear of failure in students, combining embedded sensing and actuation with a growth metaphor; used as a probe for how tangible interaction shapes reflective practice.

- **Yunnan University — Undergraduate Thesis Research** *2022 – 2023*

Recommender Systems Kunming, China

- Developed and evaluated a collaborative-filtering recommender on the goodbooks-10k dataset, comparing matrix-factorization and similarity-based approaches and analyzing recommendation quality across user cohorts.

Applied Research & Engineering Experience

- **Synexia AI** *2025 – Present*

AI Engineer (Research Intern) Ningbo, China

- Architected two deployed LLM agent systems — *Zhanlu*, a multi-tenant agent runtime, and *EDIA*, a LangGraph-based decision-intelligence system for the petrochemical sector — owning agent runtime design, model serving, evaluation, and deployment.
- **Verifiable output.** Built evaluation and guardrail infrastructure directly relevant to my research interests: golden-eval pipelines, LLM-as-judge audit gates, and citation-grounding checks that block unsourced or unverified claims from reaching users; a retrieval layer (ChromaDB, BAAI/bge-m3, hybrid SQL + vector search) enforces attribution for every generated figure.
- **Removing arithmetic from the model.** Restructured an inference pipeline so that all metrics are computed deterministically before generation, reducing end-to-end latency from 90–478 s to 30–45 s while eliminating a class of numerical hallucination — a design pattern I intend to study more formally.
- **Privacy-preserving LLM architecture.** Designed a dual-model setup in which a locally hosted model (Qwen, on-premise GPU) reads and sanitizes sensitive enterprise records before any external model sees a briefing, with external routing blocked unless the sanitization condition holds.
- Ran an MLOps loop for a multi-horizon forecasting ensemble (ARIMA, XGBoost, LSTM, STL) across 19 products, with continuous backtesting against locked baselines, measured-MAPE bias correction, and champion/challenger model selection.

- **Piliving** *June 2024 – Sept. 2024*

Technical Product Manager Ningbo, China

- Conducted a mixed-methods analysis of 100,000+ user forum posts and comments collected via the Pushshift API, applying text mining to categorize and quantify recurring product failure modes and translate them into design requirements.
- Prototyped an embedded control-panel system and parametric 3D models (Fusion 360) informed by the analysis, closing the loop from user data to physical design.

- **ProFabx** *June 2024 – Sept. 2024*

Software Engineer Ningbo, China

- Automated parametric CAD workflows through the Autodesk Inventor API in Python, reducing manual modeling time by roughly 40% and eliminating a recurring source of dimensional error.

◦ Built a browser-based parameter interface that propagates dimension changes through an entire assembly, supporting rapid design-space exploration.

Selected Technical Projects

- **InclusiveVision — Assistive Smart Glasses for Visually Impaired Users** *Dec. 2023*
  Zhejiang University [GitHub]
  ◦ Prototyped a wearable obstacle-detection system (Arduino UNO, ultrasonic sensor array) with real-time auditory feedback at up to 50 cm range, with the frame form-factor iterated through 3D-printed revisions for wearability.
- **Parametric CAD Automation via the Autodesk Inventor API** *Apr. – Aug. 2024*
  ProFabx [GitHub]
  ◦ Python-driven parametric modeling system that updates Inventor 3D models from a dimension specification, automating repetitive mechanical design tasks.
- **IoT-Based Smart Attendance System with Face Recognition** *2022*
  Yunnan University [GitHub]
  ◦ Contact-free attendance system (ESP32-CAM, on-device recognition, HTTPS cloud logging) designed to prevent proxy attendance while removing shared-surface contact.

Honors & Awards

- **United Nations Technology Bank —** Global Youth Talent Program, selected internationally among young designers*2023*

Academic Service & Teaching

- **Reviewer —** TODO: any venue you have sub-reviewed for, incl. under your advisor *Year*
- **Teaching Assistant —** TODO: course name, institution *Year*
- **Student Volunteer —** TODO: e.g. ICEC 2025 / HHME 2026 *Year*

Technical Skills

- **Programming:** Python (asyncio), C/C++, JavaScript, SQL, R , Git, Linux/Bash
- **ML & NLP:** PyTorch, TensorFlow, scikit-learn, XGBoost, Transformers (BERT, RoBERTa), sentence embeddings, fine-tuning (LoRA/QLoRA/PEFT), time-series modeling (ARIMA, STL, LSTM)
- **LLM & Agent Systems:** LangGraph multi-agent orchestration, agent runtime design, retrieval-augmented generation (hybrid retrieval, ChromaDB), prompt and context engineering, guardrails and output validation, LLM-as-judge evaluation, hallucination detection, local serving (vLLM, Ollama)
- **Research Methods:** Systematic review and screening protocols, qualitative coding, linguistic and sentiment analysis (LIWC), text mining of large social corpora, benchmarking and ablation, research through design, prototyping and user-facing evaluation
- **Systems & Data:** FastAPI, Docker, PostgreSQL/MySQL, MongoDB, Redis, Elasticsearch, ETL pipelines, NumPy/Pandas, React.js, D3.js
- **Hardware & Fabrication:** Arduino, ESP32, sensor integration and circuit prototyping, Fusion 360, Inventor API, 3D printing, laser cutting
- **Languages:** Bengali (native), English (fluent; TODO: add TOEFL/IELTS score and date), Mandarin Chinese (TODO: level; add HSK if held)